

A new multi-year data set of Potential Gradient variations at a suburban site in Northeastern Germany

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1 Abstract

The atmospheric potential gradient (PG) is a key parameter for monitoring the atmospheric electrical environment. The PG measurements are influenced by global factors, such as the global electric circuit (GEC) (Williams, 2009), as well as local weather and environmental conditions (Rycroft et al., 2000; Harrison et al., 2004). This study introduces a new PG dataset from a suburban site in northeastern Germany, a region where such recordings have been missing so far. A Campbell Scientific CS110 field mill has recorded quasi-continuously at 1-min resolution since March 2021. During validation, two systematic instrumental issues were identified and corrected: (1) a device-specific bias, addressed by statistical correction using a co-located reference field mill; and (2) humidity- and dust-driven leakage current (LC) artifacts, removed with a non-parametric regression approach. The result is a corrected 5-year record enabling detailed study of fair-weather (FW) PG variability in Central Europe.

2 Site and instrument

- Located at the Herrenkrug campus of the Magdeburg-Stendal University of Applied Sciences (52.13939°N, 11.67628°E) ~50 m a.s.l.
- CS110 field mill on a 2 m mast
- Co-located weather sensors; enclosure surrounded by a metallic fence

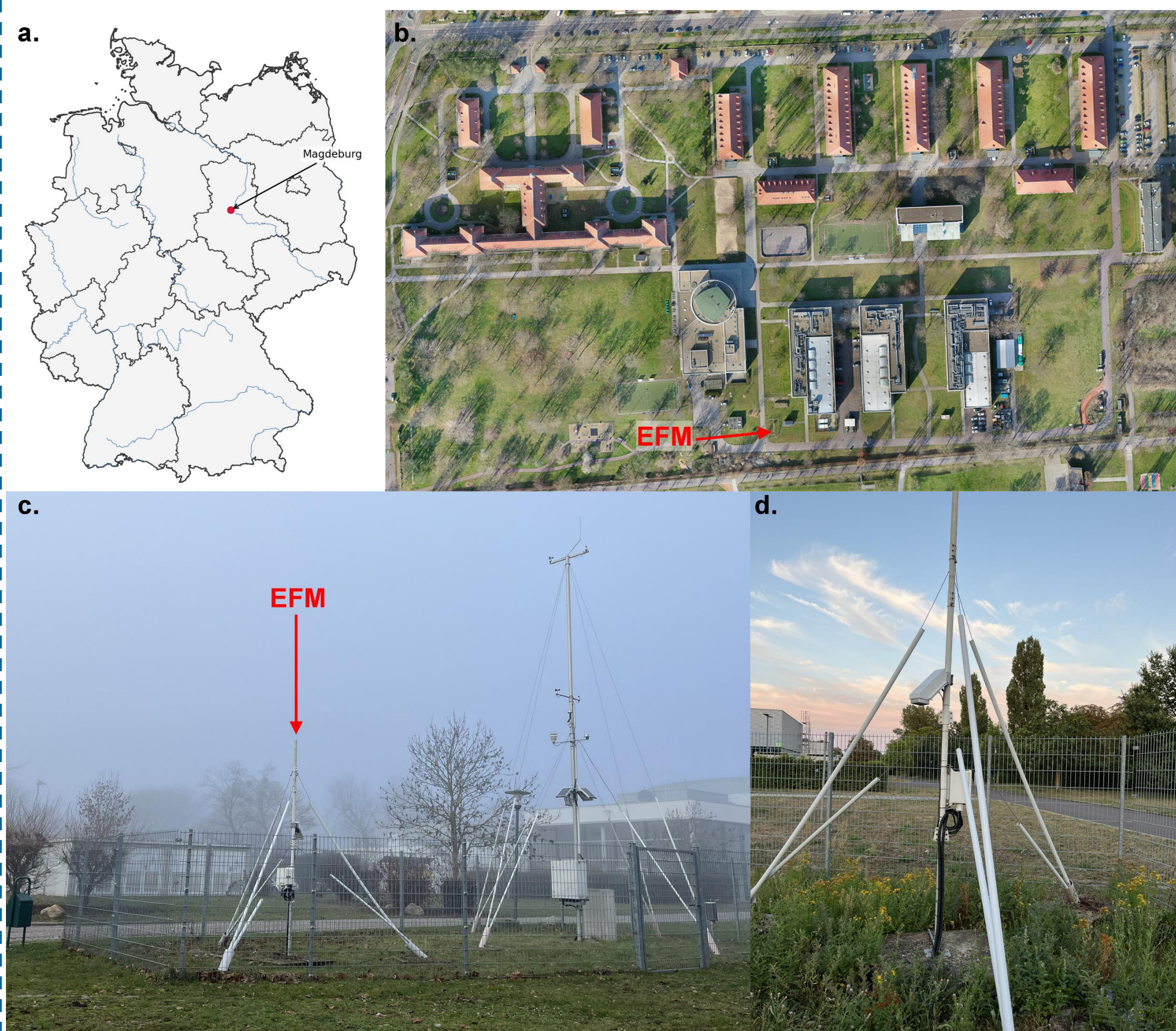


Figure 1: (a) Geographical location of the measurement site. (b) Aerial view of the Herrenkrug campus; (c) Weather sensors and field mill inside the fenced enclosure. (d) Field mill on the 2 m mast.

3 Site Correction

- Raw PG values are anomalously small: mean FW PG ~11.5 V m⁻¹ (~20 to 50 V m⁻¹), far below the global 100 -150 V m⁻¹ and shifted negative.
- 2024-2025 parallel measurements with a co-located reference field mill under FW conditions.
- Linear regression between instruments gives the correction factor:

$$PG_{corrected} = 1.25 \cdot PG_{raw} + 36.6$$

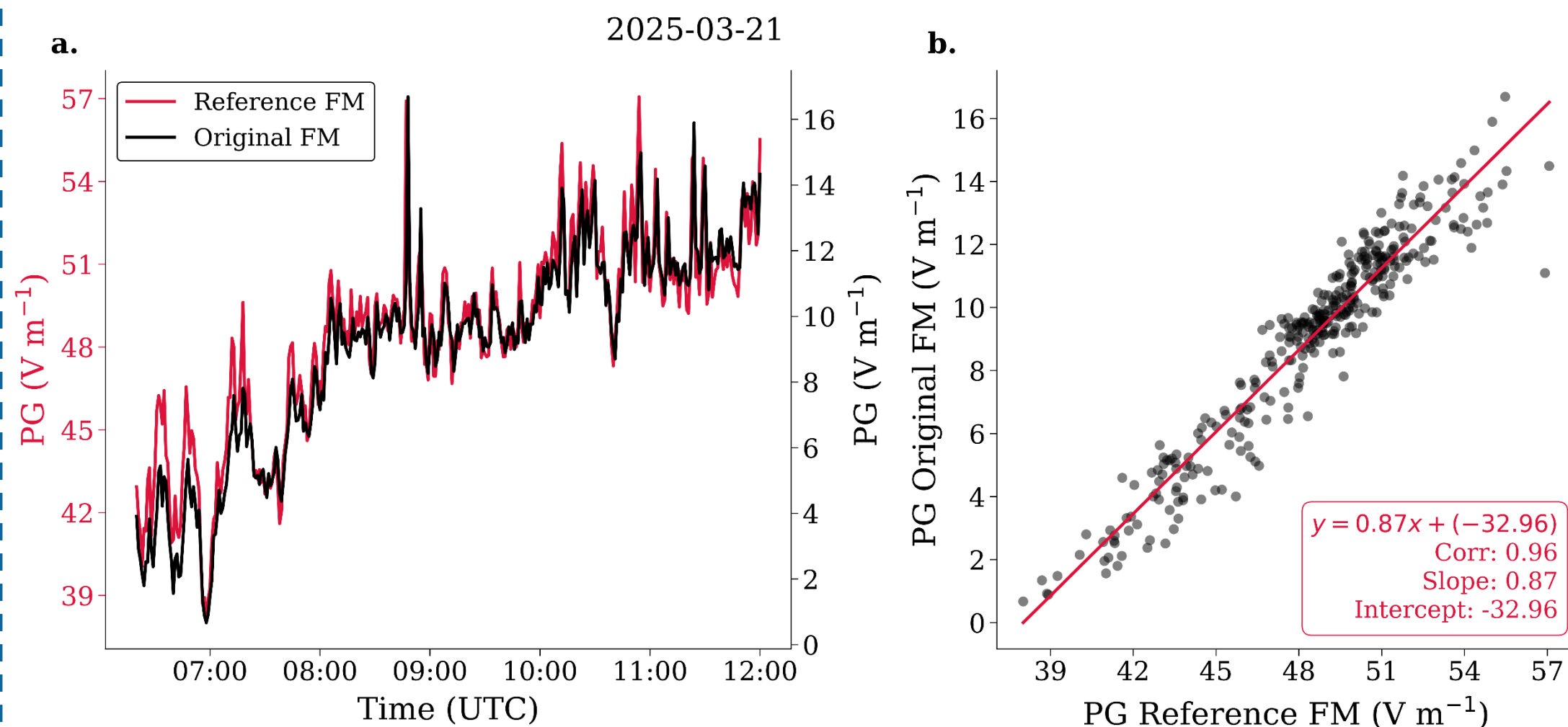


Figure 2: Parallel measurements taken on 21 March 2025 using a reference field mill on a 2 m mast inside the fenced weather-station area.

References

- Rycroft, M., Israelsson, S., & Price, C.. The global atmospheric electric circuit, solar activity and climate change. J. Atmos. Sol.-Terr. Phys., 62, 1563–1576, doi:10.1016/S1364-6826(00)00112-7, 2000.
- Harrison, R. G.: The global atmospheric electrical circuit and climate, Surv. Geophys., 25, 441–484, doi:10.1007/s10712-004-5439-8, 2004.
- Harrison, R. G.: The Carnegie curve, Surv. Geophys., 34, 209–232, doi:10.1007/s10712-012-9210-2, 2013.
- Williams, E. R.: The global electrical circuit: a review, Atmos. Res., 91, 140–152, doi:10.1016/j.atmosres.2008.05.018, 2009.

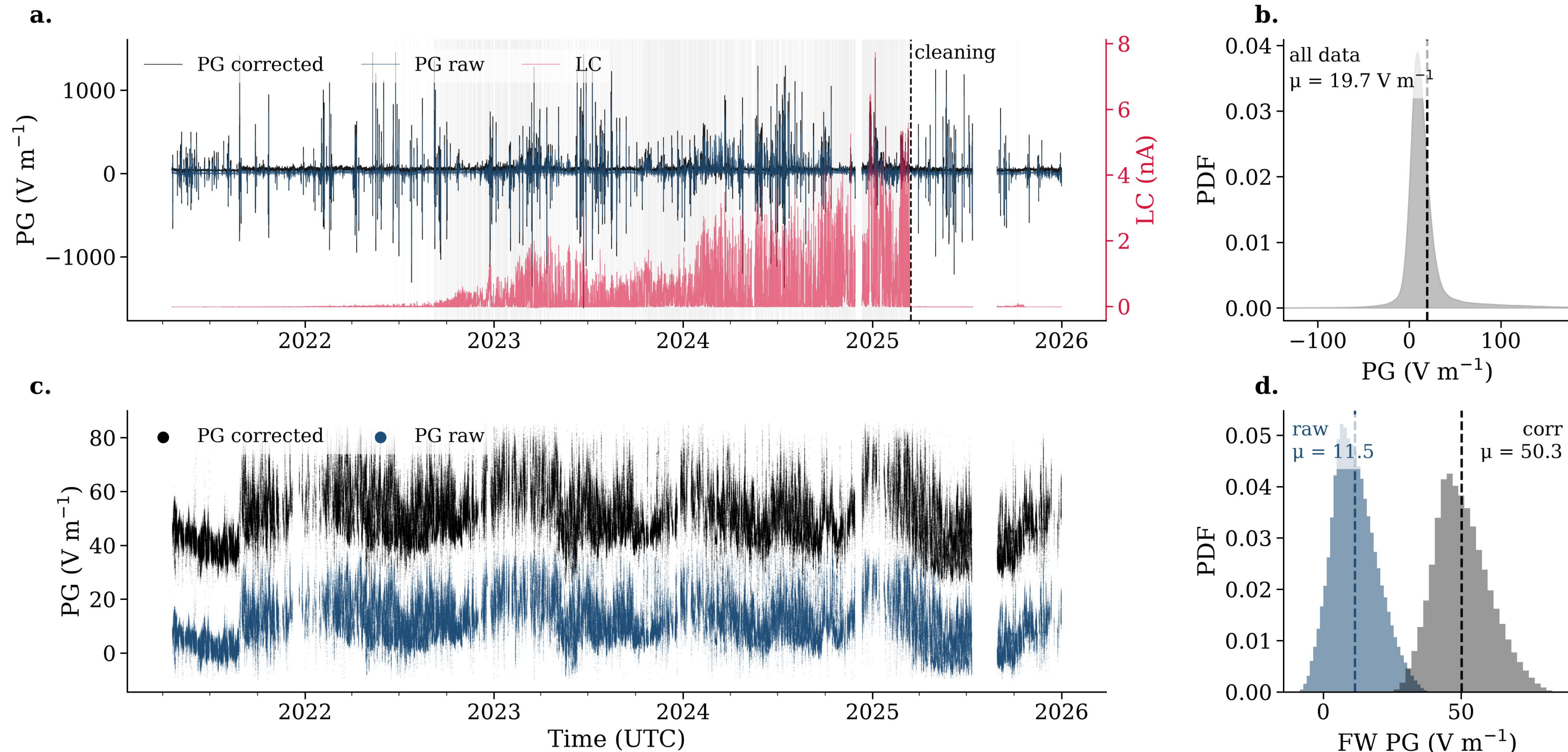


Figure 3: Time series of PG measurements from April 2021 to December 2025. (a) Raw PG (blue) and corrected PG (black) with leakage current (red); grey bands mark hourly intervals with LC > 0.05 nA. (b) PDF of all PG values. (c) Raw (blue) and corrected (black) FW PG. (d) FW PG distribution. Dashed lines in (b, d) mark the means.

4 Leakage-current artifact

- In the 2023-2024 period, humidity- and dust-driven LC distorted PG magnitude.
- LC rises after sunset and falls sharply after sunrise; the LC-PG relation is near-linear but shifts day to day.
- Contamination threshold defined at LC > 0.05 nA.
- Contaminated windows flagged by three-step validation and removed by per-window non-parametric regression.
- Instrument cleaning restored normal LC after March 2025.
- The LC artifact is not documented elsewhere.

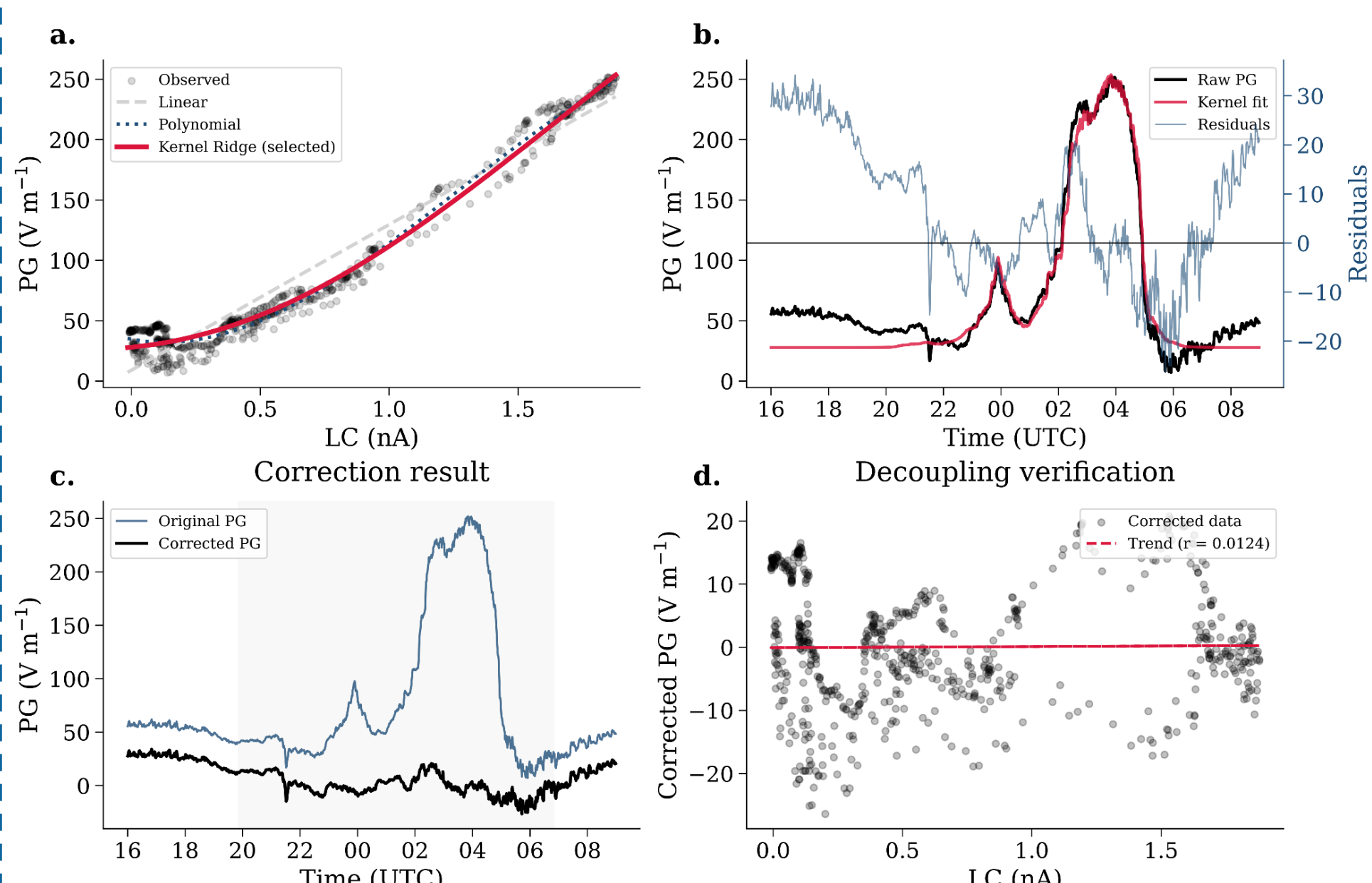


Figure 4: Identifying and removing the LC artifact. (a) Candidate regressions vs observed PG-LC. (b) Raw PG (black) and selected kernel fit (crimson); residuals (gray) are recovered PG. (c) Corrected PG (black) vs original (blue); shading marks the contamination window. (d) Verification: corrected PG vs LC.

5 Diurnal cycle

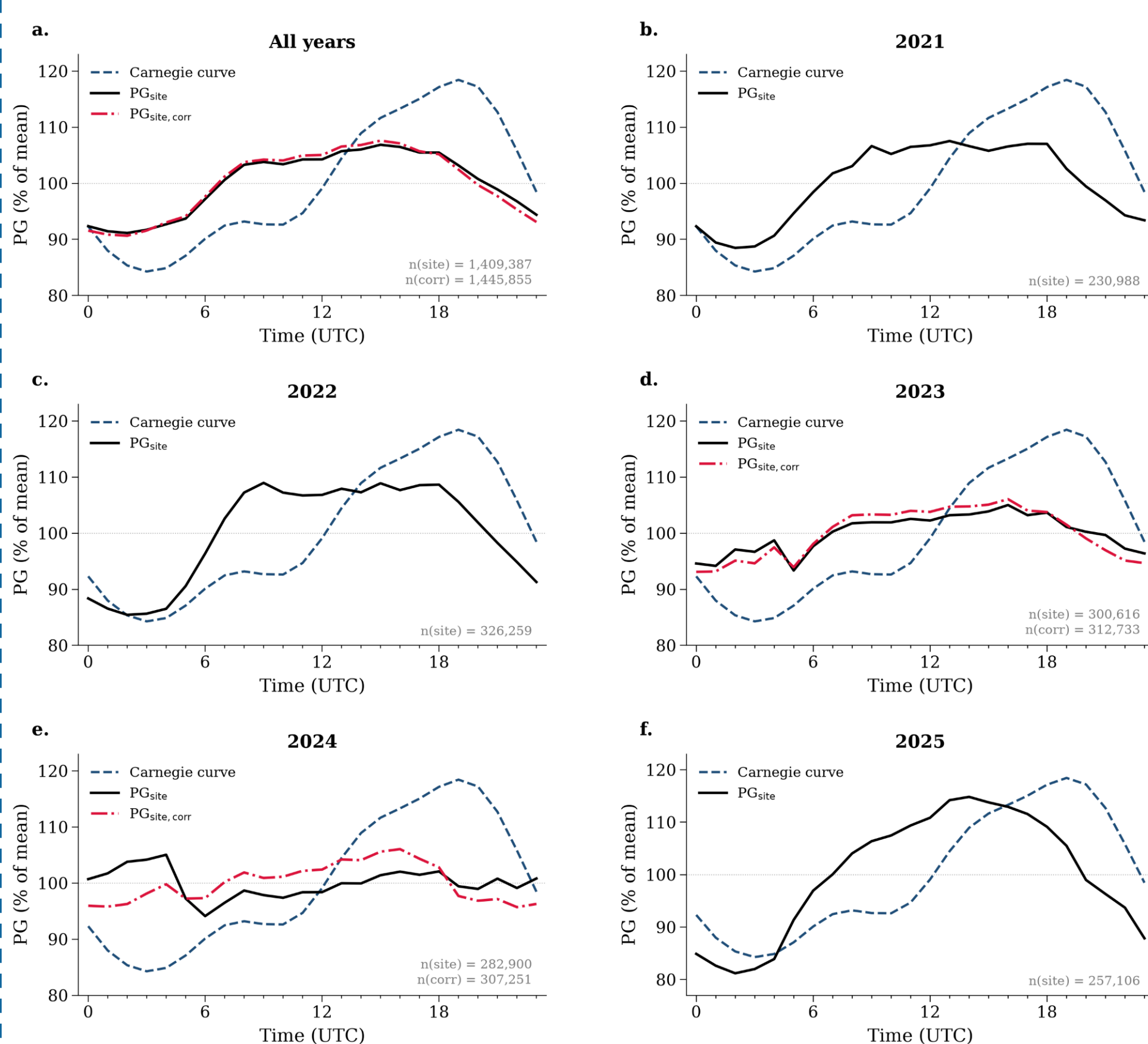


Figure 5: Median FW diurnal PG. (a) All years; (b-f) individual years. Black: final corrected PG; red dash-dot: site-corrected; blue dashed: Carnegie curve (Harrison, 2013).

Strict hourly FW selection: ≥50 valid data points, each minute with no precipitation, positive PG, RH < 85%, wind < 5 m s⁻¹, site-corrected PG 0-100 V m⁻¹.

6 Seasonal variability

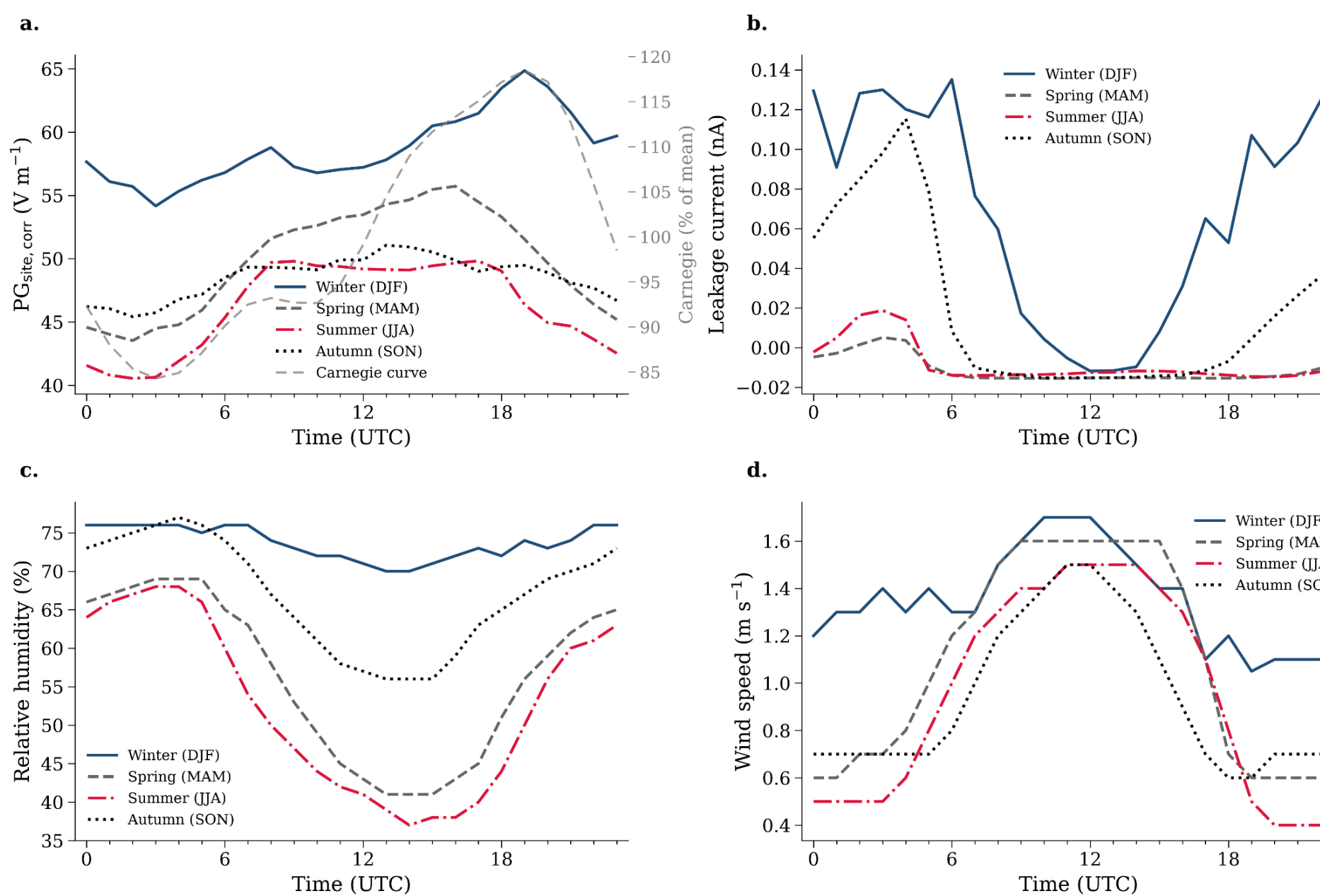


Figure 6: FW Diurnal variation of (a) PG, (b) leakage current, (c) relative humidity, and (d) wind speed

- Morning minimum in PG near 03:00 UTC in all seasons.
- Winter:** largest PG amplitude (~65 V m⁻¹), evening max at 19:00 UTC, highest RH (~76%), and overnight LC (~0.14 nA).
- Spring/Summer:** daytime maxima driven by midday RH drop; summer PG amplitude smallest, small changes in LC.
- Autumn:** Broad midday PG max, elevated early-morning LC.

7 Conclusions

- Initial PG recordings were unusually small. Several explanations were tested; after ruling out the location and atmospheric conditions, the cause was traced to a device-specific bias of recent CS110 units.
- A correction factor was derived from parallel measurements using a co-located reference field mill under FW conditions.
- LC artifact, a previously undocumented, humidity- and dust-driven leakage current, contaminates PG above 0.05 nA; we detect and remove it via per-window non-parametric regression; normal LC restored after cleaning (March 2025).
- The corrected PG recordings show a small diurnal amplitude (15.7% of the site mean, versus 34.2% for the Carnegie curve. Aside from a morning minimum near 03:00 UTC, the cycle is driven by local conditions rather than the global signal.
- Seasonal variations show the largest PG recordings in winter, and the smallest in summer; phase and amplitude are set by local aerosol and boundary-layer processes, not the global signal.

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